

Photovoltaics

```
"energy_code": "2495",  
"energy_type_name": "Photovoltaik - Photovoltaics",  
"min_power_kw": 30.005,  
"max_power_kw": 162262.2,  
"min_ref": {  
  "file_name": "EinheitenSolar_17.json",  
  "index_in_file": 46559,  
  "bundesland_code": "1402",  
  "bundesland_name": "baden_wuerttemberg",  
  "state_name_norm": "badenwuerttemberg",  
  "energy_code": "2495",  
  "energy_label": "Photovoltaik - Photovoltaics",  
  "power_kw": 30.005,  
  "commissioning_date": "2006-12-06",  
  "commissioning_year": "2006",  
  "lon": 9.956972,  
  "lat": 47.826344,  
  "google_maps_url": "https://www.google.com/maps?q=47.826344,9.956972"
```



Google Maps Distance Measured: around 37.77 m and 5.96 m





Vertical Solar Panels → 44 columns with 5 rows = 220, 220 + 4(from 45th column) = 224

Horizontal Solar Panels → 23(1st row) + 22(lowest row) = 45

Area = 0.858 m x 1.49 m ≈ **1.28 m²**

Size Calculation:

Short Side: 37.77 m/44 columns (44 solar panels vertically) = around 85.8 cm or 0.858 m

Long Side: 5.96 m/4 rows (4 solar panels at the last column) = 149 cm or 1.49 m



Left = Totally 269 (Monocrystalline or Polycrystalline?) Solar Panels

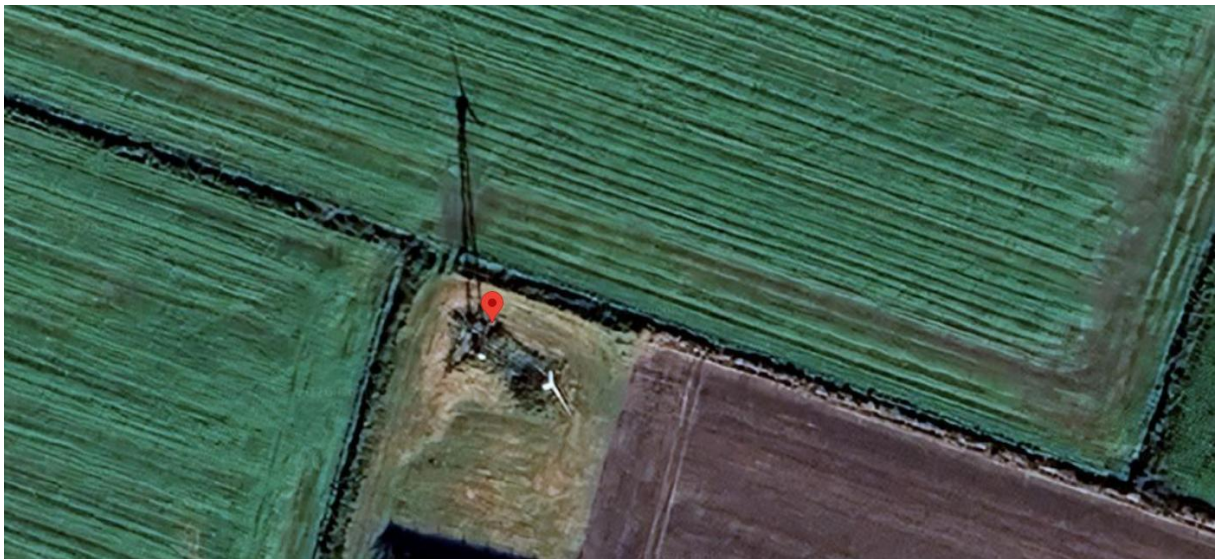
Right = 6 Solar Panels

[Complete Guide About Types of Solar Panels](#)

Since these solar panels belong to 2006, the efficiency of the solar panels can be low.

Onshore Wind Energy

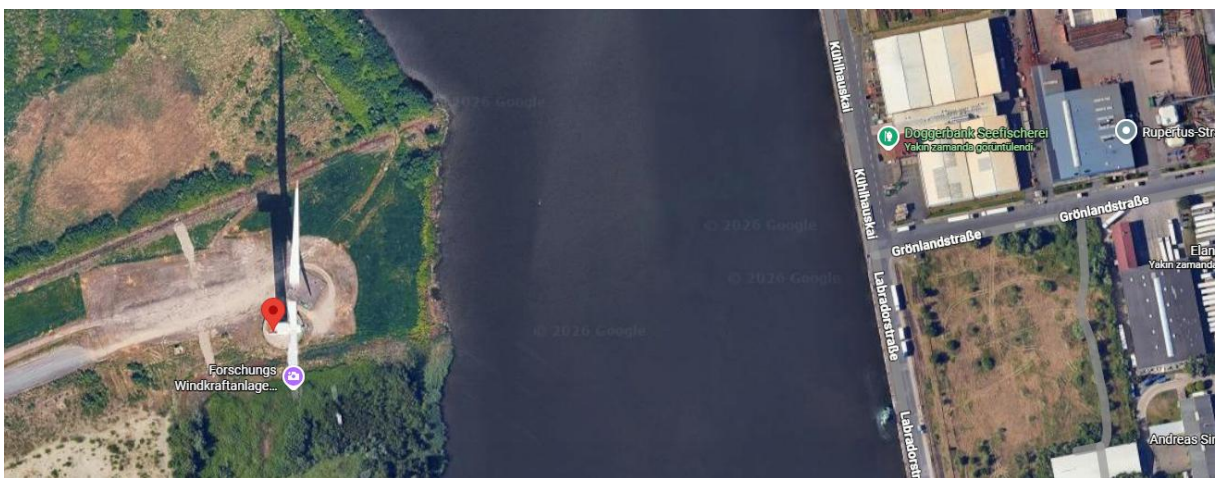
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"energy_code": "2497",  
  "energy_type_name": "Windenergie Onshore - Onshore Wind Energy",  
  "min_power_kw": 30.4,  
  "max_power_kw": 8000.0,  
  "min_ref": {  
    "file_name": "EinheitenWind.json",  
    "index_in_file": 4035,  
    "bundesland_code": "1411",  
    "bundesland_name": "schleswig_holstein",  
    "state_name_norm": "schleswigholstein",  
    "energy_code": "2497",  
    "energy_label": "Windenergie Onshore - Onshore Wind Energy",  
    "power_kw": 30.4,  
    "commissioning_date": "2016-04-29",  
    "commissioning_year": "2016",  
    "lon": 8.658601,  
    "lat": 54.543463,  
    "google_maps_url": "https://www.google.com/maps?q=54.543463,8.658601"
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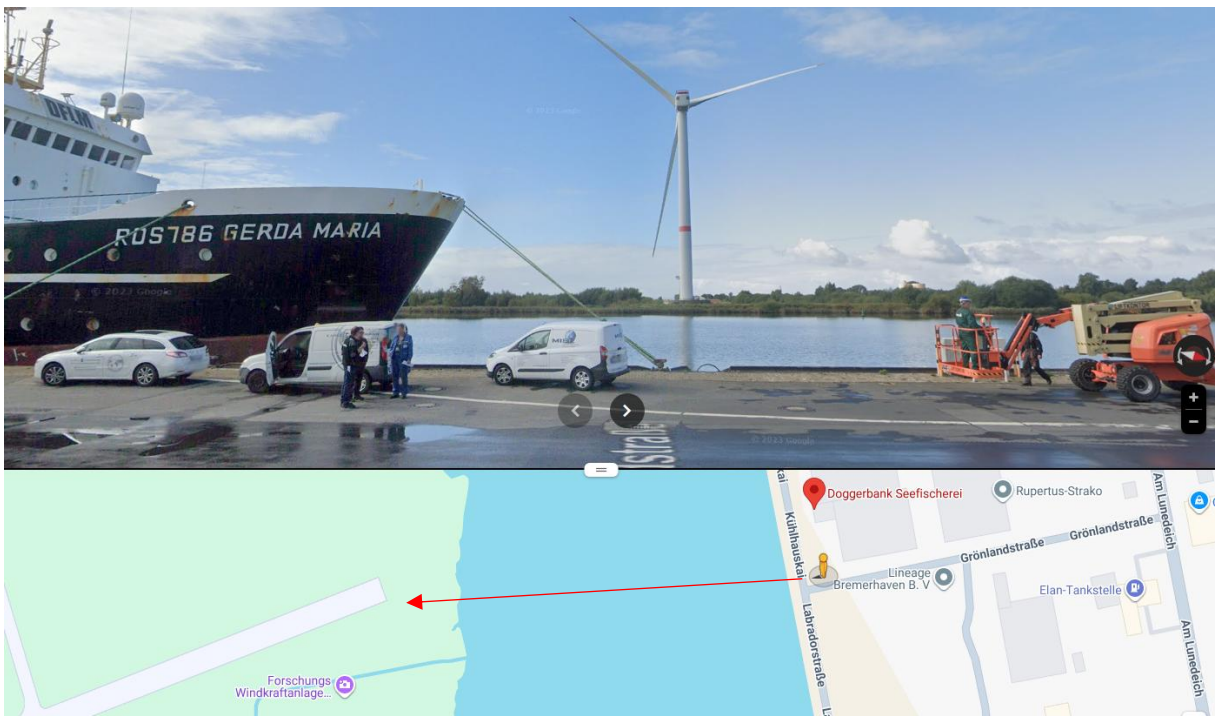
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"max_ref": {
  "file_name": "EinheitenWind.json",
  "index_in_file": 25048,
  "bundesland_code": "1404",
  "bundesland_name": "bremen",
  "state_name_norm": "bremen",
  "energy_code": "2497",
  "energy_label": "Windenergie Onshore - Onshore Wind Energy",
  "power_kw": 8000.0,
  "commissioning_date": "2017-07-25",
  "commissioning_year": "2017",
  "lon": 8.578927,
  "lat": 53.505669,
  "google_maps_url": "https://www.google.com/maps?q=53.505669,8.578927"
}

```



[Forschungs Windkraftanlage, Fraunhofer-Institut/Alfred Wegener Institut](#)



Onshore Wind Energy 2nd Maximum Value

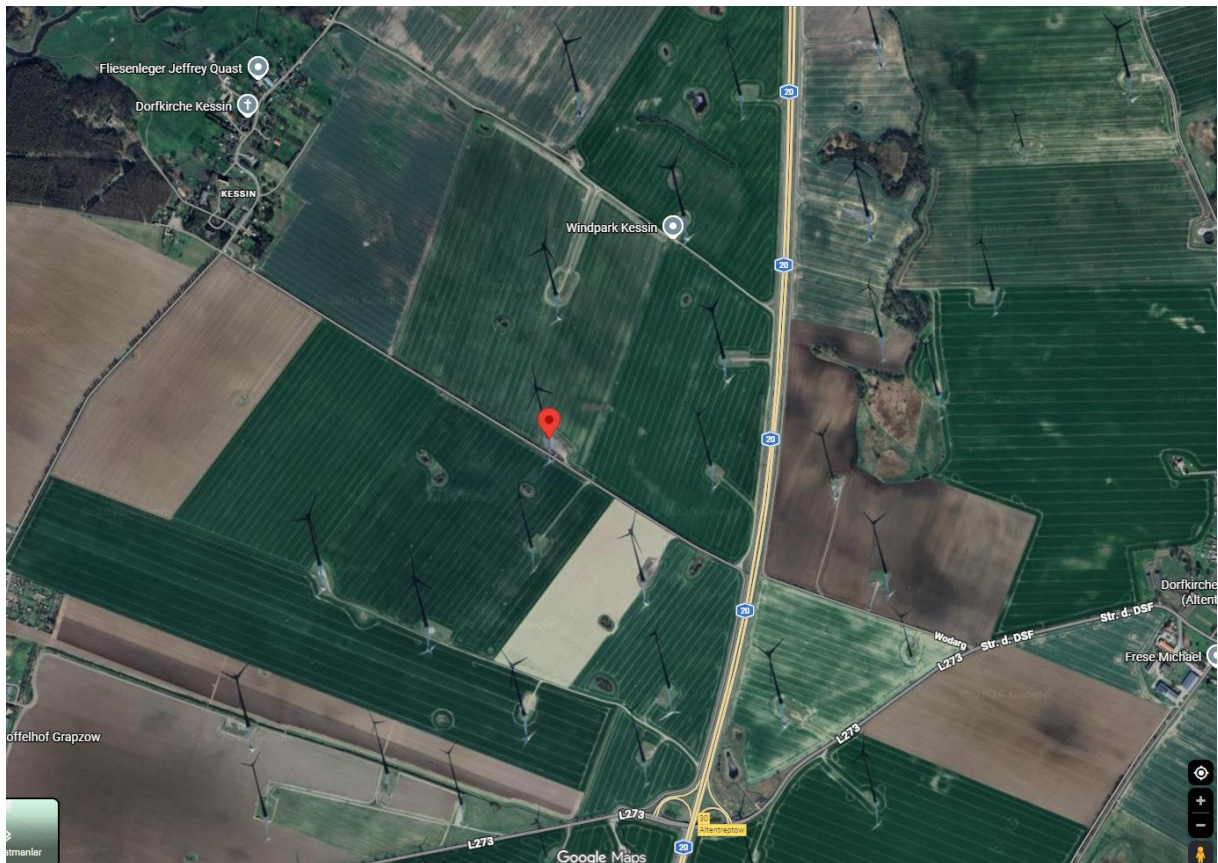
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{  
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  "index_in_file": 324,  
  "bundesland_code": "1410",  
  "bundesland_name": "rheinland_pfalz",  
  "state_name_norm": "rheinlandpfalz",  
  "energy_code": "2497",  
  "energy_label": "Windenergie Onshore - Onshore Wind Energy",  
  "power_kw": 7580.0,  
  "commissioning_date": "2012-10-29",  
  "commissioning_year": "2012",  
  "lon": 7.682222,  
  "lat": 49.978479,  
  "google_maps_url": "https://www.google.com/maps?q=49.978479,7.682222"  
}
```



→ 7.58 MW most probably belongs to these 4 onshore wind energy plants.

Onshore Wind Energy 3rd Maximum Value

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{
  "file_name": "EinheitenWind.json",
  "index_in_file": 6874,
  "bundesland_code": "1407",
  "bundesland_name": "mecklenburg_vorpommern",
  "state_name_norm": "mecklenburgvorpommern",
  "energy_code": "2497",
  "energy_label": "Windenergie Onshore - Onshore Wind Energy",
  "power_kw": 7500.0,
  "commissioning_date": "2013-09-19",
  "commissioning_year": "2013",
  "lon": 13.322525,
  "lat": 53.720963,
  "google_maps_url": "https://www.google.com/maps?q=53.720963,13.322525"
}
```



→ Here, we have a lot of onshore wind energy plants. Again power value can be total of these plants.

Onshore Wind Energy 4th Maximum Value

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{  
  "file_name": "EinheitenWind.json",  
  "index_in_file": 28830,  
  "bundesland_code": "1411",  
  "bundesland_name": "schleswig_holstein",  
  "state_name_norm": "schleswigholstein",  
  "energy_code": "2497",  
  "energy_label": "Windenergie Onshore - Onshore Wind Energy",  
  "power_kw": 7200.0,  
  "commissioning_date": "2024-08-08",  
  "commissioning_year": "2024",  
  "lon": 9.317652,  
  "lat": 54.614643,  
  "google_maps_url": "https://www.google.com/maps?q=54.614643,9.317652"  
}
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Onshore Wind Energy 5th Maximum Value

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{  
  "file_name": "EinheitenWind.json",  
  "index_in_file": 29408,  
  "bundesland_code": "1410",  
  "bundesland_name": "rheinland_pfalz",  
  "state_name_norm": "rheinlandpfalz",  
  "energy_code": "2497",  
  "energy_label": "Windenergie Onshore - Onshore Wind Energy",  
  "power_kw": 7000.0,  
  "commissioning_date": "2025-03-06",  
  "commissioning_year": "2025",  
  "lon": 7.750484,  
  "lat": 49.631822,  
  "google_maps_url": "https://www.google.com/maps?q=49.631822,7.750484"  
}
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